GMP

Another arbitrary-length number library is GMP, which is focused purely on integer mathematics. GMP is based on the GNU Multiple Precision library and is more richly featured than BCMath. GMP offers dozens of functions; let's take a look at a selection:

```
$largeNum = (string) PHP_INT_MAX;

print gmp_add($largeNum, $largeNum) . PHP_EOL;
var_dump(gmp_binomial($largeNum, 2)); // Binomial coefficient C(n, k)
var_dump(gmp_nextprime(PHP_INT_MAX));

$bin1 = gmp_init('OOO11111', 2);
$bin2 = gmp_init('11111OOO', 2);

var_dump(gmp_hamdist($bin1, $bin2)); // The Hamming distance between two
                                     binary numbers

Output:
18446744073709551614
object(GMP)#1 (1) {
  ["num"]=>
  string(38) "42535295865117307919086767873688862721"
}
object(GMP)#1 (1) {
  ["num"]=>
  string(19) "9223372036854775837"
}
int(6)
```

I've picked a few random and/or interesting functions to look at. `gmp_add()` does what it says and adds numbers. Just like BCMath, any number larger than `PHP_INT_MAX` needs to be a string or a `GMP` number class, which is returned by `gmp_init()` if you need to build one manually and generally returned by most GMP functions. GMP functions usually accept a range of types: `GMP` objects, strings, ints, etc. The `GMP` class itself is castable to a string: the return value of `gmp_add()` is a `GMP` object, but it will happily behave as a string or number in an expression.